An alternative proposal for dynamic ISF in open-source AID systems

The Diabeloop glucose curve, tested on other cohorts and set against the logarithmic and sigmoid models

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Summary

The dynamic ISF models in AndroidAPS and Trio have two origins. The logarithmic model began with Chris Wilson, who described the work in a talk to Loop and Learn in February 2022 [2]. Through an online form he asked volunteers from the diabetes online community what correction dose they would give at a stated glucose, with glucose steady and no insulin on board. About 50 people answered, from a two-year-old using 8 U a day to people in their seventies using more than 100 U, and the usable answers gave about 300 data points. The doses and their effects were reported by the respondents from experience, and neither was measured. He trained a neural network on those points and gave the surface it found across glucose and TDD as roughly 277,700 ÷ (glucose × TDD). AndroidAPS first carried that equation, from February 2022, and the glucose term was replaced by a logarithm in June 2022, which is the form in use now [3]. The talk predates that change and does not describe it.

The sigmoid model offered in Trio came from its author's observation of glucose response, and no analysis of data has been published in support of it.

Diabeloop has since examined the same idea on the records of its own closed-loop system and presented the result at the American Diabetes Association's Scientific Sessions in 2025 [1]. From 11,201 adults it reported a sensitivity of 79 mg/dL per unit at a glucose of 100 mg/dL, 35 at 180 and below 20 above 300, with a fitted equation.

This document builds on that work. It sets out the equation and how it was derived, and reports what the same measurement shows on other cohorts. It then describes how the equation could be offered in AndroidAPS and Trio as an alternative to both the logarithmic and the sigmoid models.

The measurement was repeated on 1,790 people in eight cohorts held in this series. Within person, the value at 100 mg/dL was 2.2 to 4.0 times the value at 200, against 2.61 in the published equation, so the steep fall towards 200 mg/dL is reproducible. On records where no algorithm was dosing, the published equation fits the measured values from 100 to 200 mg/dL at a single scale. On Loop records no single scale fits. Refitting the equation's five coefficients to each cohort agreed with both results, and returned a flatter curve than the published one above 200 mg/dL.

The measurement was also run on simulated people whose true sensitivity was known, and those runs are reported in full below. They describe how the measurement behaves. This document takes the published equation as given and asks how it should be scaled to a person and offered in the two systems.

The published equation has no term for the person, and it needs one before it can dose. Two are described here: the profile ISF, and total daily dose (TDD) as the logarithmic model uses. They were compared on 3,433 corrections given by 150 adults in a trial where no algorithm was dosing. The fall in glucose per unit followed the person's entered ISF, with a coefficient of 0.95 (95% interval 0.55 to 1.32) where the profile link predicts 1. Daily dose added nothing once the entered ISF was known, at -0.07 (-0.43 to 0.26) where the TDD link predicts -1. The link to the profile is therefore the candidate, and the link to TDD is kept as an arm of the evaluation.

The curve is fitted to a person by taking their profile ISF to be valid at one glucose and scaling the whole published curve to pass through it. That glucose is 140 mg/dL in this proposal, which is a modelling choice and open to revision. The publication's own reference is 180 mg/dL, and anchoring there would make every dose 1.44 times smaller. For an adult on 40 U a day with a profile ISF of 50 mg/dL per unit, the candidate gives 79.3 mg/dL per unit at 100 mg/dL, 50.0 at 140 and 30.3 at 200. AndroidAPS at an adjustment factor of 100% gives that person 43.4, 35.6 and 29.3. Both links are modelling choices made in this document and appear nowhere in the publication.

The ends of the curve sit outside Trio's default limits. Trio holds a dynamic ISF between the profile ISF divided by 1.2 and the profile ISF divided by 0.7. The candidate stays inside that band only between 109 and 158 mg/dL. Below

109 mg/dL it gives a higher sensitivity than Trio allows, reaching 1.59 times the profile ISF at 100 mg/dL against a ceiling of 1.43. Above 158 mg/dL it gives a lower one, reaching 0.30 times the profile ISF at 380 mg/dL against a floor of 0.83. With the limits unchanged, Trio would dose from the curve over 49 of the 280 mg/dL it covers. The limits need a decision from Trio's maintainers before the model can run there as published. AndroidAPS applies no such limit in its dynamic ISF calculation.

Two things are unsettled. The first is how far that comparison reaches. It covers adults on a pump who gave their own corrections, and it compares people with each other, so it does not show whether one person's value should also move with their recent insulin use, as the TDD link does. People using a loop were not tested, and nor were children. The second is where on the curve a correction is read, which moves the dose by 31 to 49%. Whether a loop dosing to this equation does better than one dosing to either existing model has not been tested.

This is a draft for discussion. The equation has not run on a device or in a closed-loop simulator, and the candidate uses the published coefficients unchanged. Readers are assumed to know how dynamic ISF works in both systems. Glucose is in mg/dL throughout because the code is.

What is the Diabeloop equation?

The source [1] is a conference abstract and poster from Diabeloop, the maker of the DBLG1 closed-loop system. It used the records of 11,201 adults with type 1 diabetes from March 2021 onwards. No full paper was found.

The abstract reports a sensitivity that rises from 65 to 79 mg/dL per unit between 50 and 100 mg/dL. Above that it falls, to 35 at 180 mg/dL and below 20 above 300. Relative to a dose sized at 180 mg/dL, the authors suggest the dose could be halved at 120 mg/dL or below and may need doubling above 340.

The poster's results figure gives the fit in two pieces, for glucose G in mg/dL. Up to 100 mg/dL:

```
ISF = 98.03 − 1.077 G + 0.008868 G^2
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Above 100 and up to 380 mg/dL:

```
ISF = 272 − 3.121 G + 0.01511 G^2 − 3.305e−05 G^3 + 2.69e−08 G^4
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The coefficients in the code (`inv008.dynisf.quartic_isf`) were checked against that figure on 11 November 2026 and match digit for digit. The quartic gives 80.6 mg/dL per unit at 100 mg/dL, 35.3 at 180, 21.1 at 300 and 15.3 at 380. It implies dose multiples of 0.56 at 120 mg/dL and 1.95 at 340, in line with the abstract's half and double. The quadratic gives 79.0 at 100 mg/dL and 65.3 at 60, so the two pieces differ by 2% where they meet.

The equation is steeper than the logarithmic model. Between 100 and 200 mg/dL its value falls by a factor of 2.61, where the logarithmic model with a divisor of 55 falls by 1.48. From 200 to 300 mg/dL its value is multiplied by 0.69 and the logarithmic model's by 0.82.

It has no term for the person. Everyone gets 35.3 mg/dL per unit at 180 mg/dL, whatever their daily dose or their profile.

How was it derived?

The unit of analysis is a fall in glucose. On days with at least 75% of time in closed loop, the authors took each glucose peak between 50 and 400 mg/dL that came at least an hour after a meal, with less than 2.5 g of carbohydrate still on board, and followed it to the next minimum more than 20 minutes later. The sensitivity for the event is the fall divided by the insulin used over it. Insulin used is the drop in insulin on board plus whatever was delivered above the reference basal. The curve is the fit through those event values against glucose.

Each event ends where glucose stopped falling, so events are selected on what glucose did. The figure is therefore a ratio of two outcomes, the fall and the insulin used during it. That bears on how the curve is read in the next section.

Four things the derivation depends on are absent from the publication. It does not define the reference basal or say which insulin action model produced the insulin on board. The figure's horizontal axis is labelled G(t) with no statement of which glucose in an event that is. It gives no daily dose for the people measured. The rest of the poster could not be retrieved, and the authors have not been asked.

What does it show on our data?

The same construction was run in this series as INV-012, on 1,790 people in eight cohorts, and is reported in section 7 of document 5. It was built before the abstract was read, and two of its event rules differ from the published

ones. It required three hours clear of a meal where the publication requires one hour with under 2.5 g on board, and a peak standing 15 mg/dL clear of its surroundings where the publication requires a 20 minute interval. Each event was filed under the glucose halfway between its peak and its minimum, and multiplied by the person's daily dose.

The shape between 100 and 200 mg/dL

The steep fall is reproducible. Within person, the value at 100 mg/dL was 2.2 to 4.0 times the value at 200 mg/dL across the eight cohorts, against 2.61 in the published equation. In REPLACE-BG, where no algorithm was dosing, the ratio was 2.55 (95% interval 2.25 to 2.82), which includes the published figure. Under closed loop it was steeper: 3.26 (3.11 to 3.39) for Loop, 3.86 (2.72 to 6.04) for oref0 and 3.21 (2.24 to 5.62) for Trio. The first two intervals lie above 2.61 and Trio's includes it.

Fitting the equation to the measured values

The table divides the measured value in each glucose bin by the published value at that glucose. The result is the daily dose, in U, at which the published equation would return what was measured. A row that stays level means the published shape fits that cohort at one scale. Only the four cohorts with carbohydrate records are shown, with the 95% interval over people.

Cohort	People	100 mg/dL	150	200	250	300
No algorithm (REPLACE-BG)	196	103 (89 to 115)	100 (94 to 108)	106 (94 to 125)	120 (103 to 137)	174 (132 to 233)
Loop	842	88 (84 to 91)	58 (56 to 60)	71 (67 to 73)	82 (77 to 89)	108 (95 to 119)
Trio	22	90 (63 to 111)	52 (40 to 64)	74 (51 to 104)		
oref0	86	109 (90 to 136)	70 (59 to 86)	78 (58 to 188)		

With no algorithm dosing, the published equation fits from 100 to 200 mg/dL at one scale: 103, 100 and 106 U, each inside the other two intervals. Above 200 mg/dL the row rises to 120 and then 174 U, so the published equation falls faster there than these records do. Under Loop no single scale fits. The figure is 88 U at 100 mg/dL, 58 at 150 and 71 at 200, and those three intervals do not overlap. Trio and oref0 show the same dip at 150 mg/dL with wider intervals.

Two cautions apply to the table. The columns hold different people, since fewer reach the higher bins, and the within-person ratios in this section are the firmer comparison. The published values were read at the centre of each bin on the assumption that the publication files an event under its midpoint glucose, which it does not state.

The scale itself is not a daily dose anyone was using. It is about 100 U in REPLACE-BG because the measurement reads high where no algorithm doses, as the simulations below show.

Refitting the equation

The equation was also refitted. The quartic, with all five coefficients free, was fitted to each cohort's values in 10 mg/dL bins, from 100 mg/dL to the highest bin holding ten people. Each bin was weighted by the square root of the people in it, which is a modelling choice, since the poster does not state its weighting. For comparison the published fit was laid over the same bins with one free number, its scale in U. The miss is the typical distance between a fit and the bin values, as a percentage of each value.

Cohort	People	Bins fitted, mg/dL	Best scale of the published fit, U	Miss, published fit	Miss, refit
No algorithm (REPLACE-BG)	196	100 to 370	103 (96 to 113)	14%	7%
Loop	842	100 to 380	71 (69 to 74)	17%	6%
Trio	22	100 to 200	65 (48 to 80)	18%	7%
oref0	86	100 to 240	82 (71 to 99)	23%	14%

The refit misses by less in every cohort, as a fit with four more free numbers must. The second table shows where the two differ. It gives each fit relative to its own value at 180 mg/dL.

Fit	People	100 mg/dL	150	200	250	300
Published		2.29	1.30	0.87	0.70	0.60
Refit, no algorithm (REPLACE-BG)	196	2.17 (1.91 to 2.40)	1.28 (1.21 to 1.33)	0.89 (0.86 to 0.92)	0.81 (0.72 to 0.87)	0.90 (0.77 to 1.02)
Refit, Loop	842	3.19 (3.02 to 3.34)	1.25 (1.22 to 1.27)	1.00 (0.98 to 1.01)	1.07 (1.03 to 1.11)	0.94 (0.88 to 1.01)
Refit, Trio	22	3.29 (2.15 to 4.76)	1.19 (0.85 to 1.53)	1.12 (0.66 to 1.61)		

Fit	People	100 mg/dL	150	200	250	300
Refit, oref0	86	2.96 (2.22 to 3.83)	1.00 (0.78 to 1.24)	1.02 (0.82 to 1.38)		

With no algorithm dosing, the refit returns the published shape from 100 to 200 mg/dL, and all three intervals there include the published figure. At 250 and 300 mg/dL the refit sits at 0.81 and 0.90 of its value at 180, where the published fit is at 0.70 and 0.60. Both of those intervals exclude the published figure. Under Loop the refit differs from the published fit at every glucose in the table. It is higher at 100 mg/dL, 3.19 against 2.29, and from 200 mg/dL upward it stays close to its value at 180. Trio's intervals, from 22 people, include the published figure at each of its three points. The oref0 values resemble Loop's, with an interval that excludes the published figure only at 150 mg/dL.

Two limits apply. A quartic suits these records poorly, because the values fall steeply to about 150 mg/dL and then level off. Under Loop the refit rises from 1.00 to 1.07 between 200 and 250 mg/dL, while the bin values themselves go from 2,182 to 2,025. Above 200 mg/dL the bins and the within-person ratios reported below are the better guide. The second limit is that the refit describes the same outcome ratio as the published curve, so the simulations in the next subsection bear on it equally.

The refit gives no single equation to offer in place of the published one. Its coefficients differ between cohorts, and its value at 180 mg/dL is 3,657 (3,431 to 4,055) with no algorithm dosing and 2,123 (2,048 to 2,214) under Loop. It confirms the published shape between 100 and 200 mg/dL on records that no loop shaped. It also agrees with the within-person ratios that, by this measure, the published fall above 200 mg/dL is not reproduced here.

What the method returns when the truth is known

The method has been run twice on simulated people whose true sensitivity was known.

In the first run (INV-012) the sensitivity was constant at every glucose, 2,000 when multiplied by daily dose, and 45% of meals went unlogged. Two arms were dosed by a simple reactive loop, at a tenth and at a fifth of its gain. They returned values 1.4 to 3.4 times too high, declining as glucose rose. Their ratio of the value at 250 to the value at 200 mg/dL was 0.87 (0.83 to 0.90) in one arm and 0.92 (0.89 to 0.95) in the other. The same loop at full gain returned 1,816 and 1,520 at 150 and 200 mg/dL, below the true value, and reached no higher bin. An arm with no loop returned 4,560, 4,509 and 6,091

at 150, 200 and 250 mg/dL. Its ratio was 1.31 (1.00 to 1.94), from 25 people, so it read high and did not fall. A constant sensitivity therefore returned a falling curve in the arms where a loop was dosing, and in the arm with none it did not.

The six Jaeb cohorts gave 0.84 to 1.05 for the same ratio, a range that contains both reduced-gain arms, so the decline in those cohorts between 150 and 250 mg/dL needs no change in what a unit of insulin does to explain it. The published equation gives 0.80, which is steeper than either arm by an amount this series cannot account for. At 100 mg/dL between 59 and 93% of the simulated events had no net insulin used, and no arm gave a value at 300 mg/dL, so the first run says nothing about either end.

The second run (INV-014) was made to cover those ends. It used TimSim, which runs the UVA/Padova model of glucose and insulin [4, 5] with added layers for day-to-day physiology, behaviour, the sensor and the infusion site. That model has no single sensitivity, so the truth was measured before the method was read. Each of 40 simulated adults was held fasting at twelve steady glucose levels, and at each a small bolus was given in one of two otherwise identical runs. The true sensitivity at that glucose is the largest gap that opened between the two, per unit given. A unit lowered glucose by 11.1 mg/dL (10.1 to 13.7) at 100 mg/dL, from 39 people, by 23.0 (16.8 to 25.9) at 200, from 39, and by 34.6 (32.0 to 51.0) at 300, from 24. Within person the value at 100 mg/dL was 0.52 (0.49 to 0.54) of the value at 200, and the value at 300 was 1.30 (1.25 to 1.34) of it. In this simulator a unit of insulin does more at higher glucose. That is a property of the model in a fasting steady state and is no evidence about people.

The same people then lived 56 simulated days under each of five dosing regimes. Two ran oref0 with SMB, one with the simulator’s infusion-site faults and one without them. Two had no algorithm and gave corrections by hand: scheduled basal and a bolus for each announced meal, with a full correction towards 140 mg/dL whenever the sensor read above 180 mg/dL, at most once in three hours. Those two differed over site faults in the same way. The fifth had scheduled basal and meal boluses, with no corrections between meals. The correction rule is a modelling choice and was not fitted to anything.

The method was run on what a real record would hold, so unannounced meals and rescue carbohydrate were absent from it. Events were filed in bins 50 mg/dL wide, where the cohorts above used 10, because 56 days gave too few events for narrower ones. Of the 40 people, 31 completed every regime,

and the table uses those 31. It gives the method's value divided by the truth, within person, and then the ratio of the value at 100 mg/dL to the value at 200 by the method and in truth.

Regime	Method ÷ truth at 100 mg/dL	At 150	At 200	100:200 by the method	100:200 in truth	People with both bins
oref0 with SMB	4.29 (3.83 to 4.50)	1.28 (1.19 to 1.40)	0.62 (0.53 to 0.73)	3.62 (3.06 to 4.30)	0.53 (0.49 to 0.55)	28
oref0 with SMB, no site faults	4.27 (4.02 to 4.61)	1.40 (1.27 to 1.51)	1.15 (0.84 to 1.24)	1.98 (1.61 to 3.02)	0.50 (0.47 to 0.61)	11
Corrections by hand	15.55 (13.47 to 20.00)	4.71 (4.00 to 5.21)	1.69 (1.35 to 1.85)	4.85 (3.92 to 6.67)	0.52 (0.49 to 0.54)	29
Corrections by hand, no site faults	17.94 (16.67 to 20.06)	5.00 (4.26 to 5.76)	1.88 (1.72 to 2.28)	4.64 (4.33 to 5.70)	0.52 (0.49 to 0.54)	29
Meal boluses only	14.90 (11.62 to 16.70)	16.00 (12.72 to 19.59)	28.18 (17.33 to 70.38)	0.24 (0.10 to 0.45)	0.52 (0.49 to 0.54)	29

In the four regimes that gave corrections, the method's value at 100 mg/dL was 2.0 to 4.9 times its value at 200, and every interval lies above 1. The truth ran the other way, at about half. The real cohorts gave 2.2 to 4.0 and the published equation gives 2.61, so the simulator returned the published shape between 100 and 200 mg/dL from a sensitivity that rose with glucose. The arithmetic behind it is the same as in the real records, described under "Near target" below. Under oref0 the median fall was 32 mg/dL for events at 100 mg/dL and 41 for events at 200, while the median insulin used rose from 0.63 to 3.88 U.

The table's values are medians over events with positive insulin used, as on the cohorts. Under oref0, 6 to 7% of events at 100 mg/dL had no net insulin used and none did at 150 or 200. With corrections by hand the share was 32 to 40% at 100 and 150 mg/dL and nil at 200.

With meal boluses only, the method's curve rose with glucose. From 150 mg/dL upward the median event in that regime had no net insulin used, and 66 to 77% of events had none. Its values rest on the remaining events, whose doses were near zero, and they ran from 15 times the truth at 100 mg/dL to several thousand times it at 300. The arm with no loop in the first run behaved

in the same way. Across both runs the curve fell where insulin was given in response to high glucose and did not fall where it was not, whatever the true sensitivity did.

The level was wrong in both directions. Under oref0 the method read 4.29 times the truth at 100 mg/dL and 0.62 of it at 200.

Above 200 mg/dL the check is thin. In the regimes with corrections the middle 90% of a typical person's sensor readings lay between 78 and 241 mg/dL. One of them, corrections by hand with site faults, held ten people at 250 mg/dL. There the method's value was 0.83 (0.75 to 1.07) of its value at 200, from 17 people, where the truth was 1.16 (1.13 to 1.22) of it. No regime with corrections filled a bin at 300 mg/dL.

Four limits apply to the second run. TimSim's own audit records that its validation figures predate a rebuild, so it is not described here as validated. The run shows what the method returns from a known truth and says nothing about what the truth is in people. The truth was measured fasting with the added layers off, and a free-living high usually has carbohydrate still absorbing and more insulin on board. A truth measured inside the free-living runs was not attempted. The bins are five times wider than those used on the cohorts. The nine people left out each had a run that ended early, so the 31 are selected on having completed. Nine runs ended through hypoglycaemia, eight of them under oref0, and four through hyperglycaemia, all with meal boluses only.

Near target

Near target the decline comes from the insulin, with the fall almost unchanged. Across cohorts the median fall was 30 to 43 mg/dL for events at 100 mg/dL and 39 to 64 for events at 200. The median insulin used was 0.02 to 0.2 U at 100 mg/dL and 0.3 to 1.5 U at 200. A similar fall divided by much less insulin gives a much higher figure. At 100 mg/dL between 9 and 58% of events had no net insulin used at all, and a line fitted through each person's events put the fall at zero net insulin at 27 to 44 mg/dL. In type 1 diabetes a raised glucose does not fall without insulin, so that fall is basal and earlier doses acting, together with the selection of events that end at a minimum. The steep part of the curve below 150 mg/dL therefore rests on very small net doses. The second simulation showed the same pattern, and there the value at 100 mg/dL was 4.3 to 17.9 times the truth.

Above 200 mg/dL

The two measures disagree here. By the ratio of fall to insulin used, the value at 300 against 200 mg/dL within person was 0.90 (0.77 to 1.26) in REPLACE-BG, from 60 people, and 0.96 (0.84 to 1.08) under Loop, from 137 people. Both intervals exclude the published 0.69. Measured as the extra fall per extra unit, the same comparison gave 0.66 (0.49 to 0.73) in REPLACE-BG, from 24 people, and 0.73 (0.64 to 0.93) under Loop, from 49 people, and both of those include 0.69. No simulated regime with corrections reached 300 mg/dL, so neither measure has a known-truth check there. The extra fall per extra unit was not computed in the second simulation.

As a predictor

The shape has been tried once as a predictor on oref records, where it did worse than the static profile. On 80 people and 62,344 overnight windows, with each shape anchored to the profile at 100 mg/dL, the typical miss was 24.2 mg/dL for the static profile and 29.8 for this shape. At 205 to 260 mg/dL it was 37.8 against 73.7. No interval was reported for those figures. Two things limit the test. Its windows were selected on the later outcome, as the audit of 16 September 2026 recorded, and it anchored the shape at 100 mg/dL where this document anchors at 140.

Where that leaves the equation

The publication is a conference abstract from the system's manufacturer, drawn from the records of 11,201 adults, and this document takes its equation as given. The same measurement on other people's records reproduces it between 100 and 200 mg/dL. Above 200 mg/dL the records here are flatter than the published fit by one measure and agree with it by the other.

The simulations describe how the measurement behaves on constructed records. They say nothing about what the truth is in people, and they are reported so that a reader can weigh them.

Two questions remain for this series. One is the scale for each person, taken up in the next section. The other is whether a loop that doses to the equation does better than one dosing to either existing model or to a flat profile. That has not been tested, and the evaluation proposed below is designed to answer it.

A proposal from June 2026 that used this shape (document 3) was withdrawn for live dosing on 16 September 2026. The audit of that date left the way open for such candidates to be simulated and replayed in shadow, and to be studied prospectively. The evaluation proposed below stays inside that.

How could it be implemented?

What it would replace

AndroidAPS and Trio run the same logarithmic equation:

sensitivity = 1800 ÷ (adjustment factor × TDD × ln(glucose ÷ divisor + 1))

Trio computes a ratio, profile ISF × adjustment factor × TDD × ln(glucose ÷ divisor + 1) ÷ 1800, and divides the profile ISF by it, so the profile value cancels. Given the same inputs the two systems return the same number. This was checked on 11 November 2026 against AndroidAPS dev 0c593a4b09 and Trio dev 902be75e8, both of 10 November 2026. Neither calculation differs from the released versions (AndroidAPS master f77abd9b82, Trio main 0b29cd534), and Trio's `DynamicISF.swift` was last changed on 24 August 2026.

What separates the two in use is the defaults around the equation. Trio's adjustment factor defaults to 0.8 where AndroidAPS defaults to 100%, which puts Trio's value 25% higher. Trio then holds its ratio inside the autosens limits, 0.7 and 1.2 by default, and `calculateRawDynIsf` in AndroidAPS applies no limit. For the worked person Trio's upper limit binds from 157 mg/dL. Above that glucose its value stays at 41.7 mg/dL per unit while AndroidAPS continues down to 29.3 at 200 mg/dL. The daily dose is blended differently in each system, and the divisors differ for faster insulins: 55, 65 or 75 by peak time in AndroidAPS, and 55 or 70 in Trio, or 120 less a custom peak time.

Trio's sigmoid mode returns the profile ISF at the profile target and moves it within the same limits. oref0 as published has autosens and no dynamic ISF based on daily dose. Loop and iAPS were not read.

Linked to the profile ISF, or to TDD?

The published values need a scale for each person. At 140 mg/dL the equation gives 50.9 mg/dL per unit to a person on 20 U a day and to a person on 80 U a day. Among 422 adults with entered pump settings analysed earlier

in this series, sensitivity multiplied by daily dose was 1915 (95% CI 1828 to 1986), which puts the first of those people near 96 mg/dL per unit and the second near 24.

The approach is to titrate the curve to the person. Their entered setting is taken to be valid at one glucose, the anchor, and the whole published curve is multiplied by the one number that makes it pass through that value. The shape is left as published. The curve moves up or down for each person, and the ratio between its values at any two glucose levels stays as published.

The anchor in this proposal is 140 mg/dL. That is a modelling choice, made as the initial proposal, and nothing in the publication or in the analyses here selects it. The publication's own reference point is 180 mg/dL, the glucose against which it states its halving and doubling of doses, and earlier drafts of this document anchored there. The published value at 140 mg/dL is 1.44 times the value at 180. Anchoring at 140 therefore gives every person a sensitivity 1.44 times lower at every glucose than anchoring at 180 would, and a dose 1.44 times larger.

Linked to the profile, the entered ISF is taken to hold at 140 mg/dL:

sensitivity = profile ISF × (published value at this glucose ÷ published value at 140 mg/dL)

This fits Trio's structure, which already expresses dynamic ISF as a ratio of the profile. The schedule through the day is kept and no level has to be chosen. The value does not respond to recent dosing. It is also only as good as the entered number. It suits a profile ISF the person has evaluated, and it carries the whole of any error in one they have not.

Linked to TDD, a person's sensitivity at 140 mg/dL is taken to be 1915 ÷ TDD:

sensitivity = (1915 ÷ TDD) × (published value at this glucose ÷ published value at 140 mg/dL)

This is the logarithmic model with its glucose term swapped. The daily dose is each system's existing blend and the adjustment factor multiplies it as now. The divisor for insulin type has no counterpart and drops out. The value keeps moving with recent insulin use by the same multiplier as today: 1.39 for a person whose blended dose is 0.72 of their own average and 0.88 at 1.14 of it, which are the tenth and ninetieth percentiles among 960 people in document 6. The profile ISF and its schedule through the day play no part, as

in AndroidAPS now. The adult figure of 1915 is a modelling choice. The figure pooled across all ages in the same analysis is higher, and 51% of that group are children, so it was not used.

The two links agree for anyone whose profile ISF multiplied by daily dose is 1915. They differ for everyone else, and that difference was used to test them.

Which link corrections followed

The test (INV-015) used REPLACE-BG, the one cohort here in which a correction was the person's own dose and no algorithm added to it. A correction counted when it was given at 150 to 350 mg/dL and was at least 0.5 U, with no carbohydrate entered for six hours either side and no further bolus for six hours. That left 3,433 corrections from 150 adults who each had five or more. Among them the entered ISF multiplied by daily dose had a median of 1929. A tenth of people sat below 1345 and a tenth above 2997, and that spread is what separates the links.

For each person the fall in glucose over six hours, per unit given, was divided by the published shape at the starting glucose, and the median was taken. That figure is a ratio of outcomes and only its ordering of people was used. It was related to daily dose and to the entered ISF together, in logarithms. The expectation written before the run was that the result would lean towards the profile.

	Coefficient on daily dose	Coefficient on entered ISF
The link to TDD predicts	-1	0
The link to the profile predicts	0	1
Measured (95% interval)	-0.07 (-0.43 to 0.26)	0.95 (0.55 to 1.32)

Both measured intervals include what the link to the profile predicts and exclude what the link to TDD predicts.

The second table puts the same people in four groups of equal size by their entered ISF multiplied by daily dose. The first group entered an ISF well below what their daily dose would suggest, and the last group one well above it. The last column is the fall achieved per unit as a multiple of the entered ISF.

Entered ISF × daily dose	People	Start, mg/dL	Target	At six hours	Achieved ÷ entered ISF (95% interval)
900–1665	38	209	106	135	1.38 (1.23 to 1.76)

Entered ISF × daily dose	People	Start, mg/dL	Target	At six hours	Achieved ÷ entered ISF (95% interval)
1665–1929	37	240	110	130	1.76 (1.61 to 1.92)
1929–2322	37	234	110	142	1.50 (1.29 to 1.68)
2322–5211	38	229	107	138	1.47 (1.40 to 1.63)

All four groups stood at 130 to 142 mg/dL six hours after a correction. The last column is 1.38 in the first group and 1.47 in the last, with the second group highest. Under the link to TDD it would fall as the product rises.

The comparison between groups does not depend on the anchor, and nor do the coefficients above. The level of the column does. With the anchor at 140 mg/dL all four intervals lie above 1, so the six-hour fall per unit was 1.38 to 1.76 times the sensitivity the candidate assigns at the starting glucose. With the anchor at 180 mg/dL the same column read 0.96 to 1.22. The fall includes basal and earlier doses acting, so the column overstates what the correction did and cannot place the anchor. It is the one figure in this document in which the anchor meets a measured fall, and it is reported for that reason.

The test has a known lean. A correction is sized from the entered ISF, so any part of the fall that does not depend on the dose makes the entered ISF look like the better scale. Falls were therefore constructed for the same people and doses with each link as the truth, and the test was run on them. The two cases in which the fall follows the distance from target were added after the first run had returned a coefficient near 1 on the entered ISF. With the link to TDD as the truth and nothing else acting it returned -0.97 on daily dose and -0.05 on the entered ISF, close to that link’s prediction. With 46% of the fall following the distance from target whatever the dose, it returned -0.54 (-0.68 to -0.42) and 0.37 (0.22 to 0.51), and both values lie outside the measured intervals. The 46% is the fall on nights with no bolus as a share of the fall per unit on correction nights in this cohort (INV-009). At 75%, which is a modelling choice with no measurement behind it, the constructed values came inside the measured intervals. With the link to the profile as the truth, the test returned 0.88 to 0.97 on the entered ISF in every case. The link to TDD is therefore excluded as the scale between people unless about three quarters of the six-hour fall was independent of the dose.

The result has limits. It comes from 150 adults in one trial. It compares people with each other, so it does not test whether one person's sensitivity moves with their recent dosing, which is what the link to TDD adds within a person. Document 6 measured that movement at an exponent of -0.645 (standard error 0.033) across 1,313 people. People using a loop were not tested, because in their records the algorithm doses alongside the correction, and children were not tested. The 46% was measured overnight and the corrections were given at any hour.

One person under each model

The worked person has a profile ISF of 50 mg/dL per unit, a daily dose of 40 U, rapid-acting insulin and recent dosing equal to their average. Their product is 2,000, which is why the two links give them values about 4% apart. The table gives the sensitivity each method assigns them, in mg/dL per unit, with both systems at their defaults.

Glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Trio sigmoid	Equation linked to TDD	Equation linked to profile
80	50.1	62.6	53.7	75.9	79.3
100	43.4	54.3	50.0	75.9	79.3
120	38.9	48.6	47.1	59.8	62.5
140	35.6	44.4	45.0	47.9	50.0
150	34.2	42.8	44.3	43.2	45.1
180	31.0	41.7	42.9	33.2	34.7
200	29.3	41.7	42.4	29.0	30.3
250	26.3	41.7	41.8	23.3	24.4
300	24.1	41.7	41.7	19.9	20.8
380	21.8	41.7	41.7	14.4	15.0

Against AndroidAPS at 100% the equation linked to the profile gives this person less insulin below 211 mg/dL with a divisor of 55, and below 168 mg/dL with a divisor of 75, and more above. The ratio between the two sensitivities is 1.83 at 100 mg/dL, 1.32 at 150, 1.03 at 200 and 0.86 at 300. Against Trio at its defaults it gives less insulin below 158 mg/dL in logarithmic mode and below 152 in sigmoid mode, and more above. Against the person's own profile it gives less insulin below 140 mg/dL and more above, which holds for everyone. The comparison with AndroidAPS holds for this person only, since AndroidAPS scales by daily dose and the candidate by the profile.

The published fit stops at both ends, and the reference implementation (`inv013/candidate.py`) makes a choice at each. Below 100 mg/dL it holds the value reached at 100. The published quadratic falls there, towards 65 mg/dL per unit at 60 mg/dL, and following it would give more insulin as glucose drops below 100. Work in this series in March 2026 declined the quadratic for dosing on that ground. Above 380 mg/dL, where the fit ends, the value at 380 is held.

This document carries the link to the profile as the candidate, on the result above, and the link to TDD as an arm of the evaluation.

The ends of the curve and Trio's limits

Trio holds the ratio of the profile ISF to the dynamic value inside the autosens limits, 0.7 and 1.2 by default. The candidate is the profile ISF multiplied by the curve's shape, so the limits act on the shape itself: it may run from 0.83 to 1.43 of its value at 140 mg/dL. The published shape runs from 0.30 to 1.59. It lies inside the limits between 109 and 158 mg/dL and outside them over the rest of its range. Those two glucose values are the same for every person on the profile link, whatever their profile ISF.

The table gives the worked person's sensitivity from the curve, and the value Trio's default limits would pass on to the dose, in mg/dL per unit.

Glucose, mg/dL	Curve linked to profile	With Trio's default limits	Held by
100	79.3	71.4	Lower limit, 0.7
120	62.5	62.5	
140	50.0	50.0	
150	45.1	45.1	
180	34.7	41.7	Upper limit, 1.2
200	30.3	41.7	Upper limit, 1.2
250	24.4	41.7	Upper limit, 1.2
300	20.8	41.7	Upper limit, 1.2
380	15.0	41.7	Upper limit, 1.2

The two ends are affected unequally. Near target the curve withholds slightly more insulin than the limits allow: at 100 mg/dL it asks for 0.63 of what the profile would, and the lower limit holds that at 0.7. At high glucose the gap is wide. The curve asks for 2.41 times the profile's insulin at 300 mg/dL and 3.33

times at 380, where the upper limit holds 1.2. A correction from 300 mg/dL to 100, read at the starting glucose, is 9.62 U from the curve and 4.80 U inside the limits.

With the limits as they are, the model in Trio would follow the curve between 109 and 158 mg/dL and give 1.2 times the profile's insulin at every glucose above that, so the published curve would not be what doses over most of its range. Running the curve as published needs limits for this mode of about 0.63 and 3.33, or no limits on it, and the upper figure is the one that carries risk, since it more than triples a correction at the top of the range. Which of these Trio should do is a question for its maintainers. The third stage of the evaluation runs the candidate with and without the default limits so that the decision has figures behind it. `calculateRawDynIsf` in AndroidAPS applies no limit, so the whole curve would reach the dose there, and the same question about the top of the range applies to it.

The level of the TDD arm

The link to the profile needs no level, because the entered ISF supplies it. The link to TDD needs one, and the publication gives nothing to set it from. One way to compare the possible levels is the daily dose at which the published values would apply unchanged, called the reference dose here.

Basis for the level	People	Sensitivity × daily dose at 200 mg/dL (95% interval)	Reference dose, U
Adult entered settings, taken to hold at 140 mg/dL (the TDD arm)	422	1,161 (1,108 to 1,204)	37.7
Adult entered settings, taken to hold at 180 mg/dL (the publication's reference)	422	1,674 (1,598 to 1,736)	54.3
The 1800 rule, taken to hold at 105 mg/dL (this series, March 2026)		731	23.7
Measured under Loop	842	2,182 (2,065 to 2,266)	70.8
Measured under Trio	22	2,273 (1,569 to 3,218)	73.7

Basis for the level	People	Sensitivity × daily dose at 200 mg/dL (95% interval)	Reference dose, U
Measured under oref0	86	2,398 (1,779 to 5,809)	77.8
Measured with no algorithm dosing (REPLACE-BG)	196	3,283 (2,898 to 3,854)	106.5

The first row takes a person's entered sensitivity to be right at 140 mg/dL, and the second at 180. The third takes the conventional rule to be right near target, and it puts the whole curve lower again. The last four rows are the measured values from the previous section, read at 200 mg/dL. All four intervals lie above the arm's 1,161. Trio's, from 22 people, includes the 1,674 of the second row, and the other three lie above that as well. How far the measured rows sit from a true level is not known. At 200 mg/dL the same measurement on simulated people read between 0.62 and 2.4 times the truth across the seven regimes that dosed in response to glucose, and daily dose was not defined the same way in each source.

No default can be chosen from these figures. The reference doses run from 24 to 107 U. The measurement cannot narrow that range alone, given how far it sat from the truth in simulation, so the arm's level is left to the fourth stage of the evaluation, which gives each person a starting adjustment factor. For any one person the adjustment factor moves the level in either direction, as it does now. Its guidance has to be worked out again for this arm, because at 100% the arm's value at 100 mg/dL is 1.75 times the existing one.

Where on the curve a correction is read

A correction from 200 mg/dL to a target of 100 passes through every glucose in between, and the equation gives a different sensitivity at each. The existing code reads the logarithmic model at the current glucose, or at a glucose chosen by one of three branches in `DetermineBasalSMB.kt`. This draft assumes the publication filed each event under its midpoint glucose, and that assumption has not been confirmed.

The table gives the units needed to bring the worked person to 100 mg/dL under each reading, with the equation linked to the profile. It is the arithmetic of the sensitivity alone, before insulin on board and the loop's predictions change it.

Starting glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Profile	Equation, read at the start	Equation, read at the midpoint	Equation, summed along the path
150	1.46	1.17	1.00	1.11	0.85	0.86
200	3.41	2.40	2.00	3.30	2.22	2.24
250	5.71	3.60	3.00	6.15	4.16	4.10
300	8.29	4.80	4.00	9.62	6.60	6.32

Reading the equation at the starting glucose asks for 31 to 49% more insulin than reading it at the midpoint. From 200 mg/dL the two readings give 3.30 and 2.22 U, against the 2.00 U the flat profile gives. Adding up the equation along the whole path gives 2.24 U. The midpoint reading stays within 0.06 U of that sum up to a starting glucose of 250 mg/dL and is 0.28 U above it at 300.

The default proposed is the midpoint between the glucose being corrected and the target, which is a modelling choice. With the anchor at 140 mg/dL, that reading gives a correction from 180 mg/dL to a target of 100 the dose the profile ISF alone would give. Reading at the starting glucose is kept as an arm of the evaluation.

The anchor and the reading together set how much insulin the candidate asks for overall. Each form was replayed on the recorded loop decisions of 138 people (INV-016), of whom 22 used Trio, 19 AndroidAPS and 97 oref0. The insulin asked for at every reading above 100 mg/dL was summed and set against the same sum under the profile ISF alone. Read at the midpoint, the candidate asked for 1.00 (95% interval 0.94 to 1.03), 1.00 (0.95 to 1.05) and 1.03 (1.01 to 1.06) times the profile's correction insulin in those cohorts. Read at the starting glucose it asked for 1.39 (1.26 to 1.44), 1.38 (1.29 to 1.47) and 1.44 (1.39 to 1.51) times. An anchor of 180 mg/dL lowers every one of those figures by the factor of 1.44. The replay shows what each form would have asked for at readings recorded under another algorithm, and says nothing about where glucose would then have gone.

What changes in the code

The glucose function is the smallest part. One function per system returns the published quartic divided by its value at 140 mg/dL, with test vectors shared between the two codebases so that both return the same values.

Trio's limits need a decision from its maintainers, as set out above: with the defaults unchanged the curve reaches the dose only between 109 and 158 mg/dL.

The forecasts are affected as well as the dose. Both systems calculate a sensitivity at each predicted glucose, and the call sites can stay as they are. The equation's value rises 2.61 times as predicted glucose falls from 200 to 100 mg/dL, against 1.48 for the logarithmic model, so a forecast heading towards target is computed with a higher sensitivity than now. The effect on minimum predicted glucose and on time spent at zero basal has not been measured and is part of the replay.

The remaining additions are smaller. The loop result already stores the sensitivity it used (`variableSens` in AndroidAPS), and during evaluation both the existing and the new value are stored while the existing one doses. Settings gain a mode selector and, in a first version, no new number for the person to set. Each system gets unit tests on the shared reference vectors and replay assets for the new mode, alongside its existing tests (`ReplayApsResultsTest` in AndroidAPS, `DynamicISFTests` in Trio). A guard test confirms that the existing modes' output is unchanged.

In AndroidAPS the code touched is `calculateRawDynIsf` and the preference screen, together with the per-prediction lines and the dose-sensitivity branches in `DetermineBasalSMB.kt`. In Trio it is `DynamicISF.swift` and the forecast adjustment used in logarithmic mode. The link to the profile brings the profile ISF into the AndroidAPS calculation, which does not use it now.

Evaluation

Eight arms are compared throughout. Three are references. The logarithmic model as shipped and Trio's sigmoid are the two existing models, and the profile alone is the baseline for both. Four arms are forms of the equation: linked to the profile and read at the midpoint, the same read at the starting glucose, linked to TDD, and linked to a plain average daily dose. The eighth is the first draft of this document, a sensitivity anchored to the profile and flat above 100 mg/dL. It is the arm the equation has to beat for its shape to matter.

The first stage replays each arm's value on recorded histories using the INV-008 pipeline. Results are split by glucose and by time since the last meal, and reported separately for people who announce meals and people who do

not. The stage shows how much insulin each arm asks for and in whom. It is descriptive and has no pass rule. It was run on 138 people as INV-016, and its totals for the candidate are quoted above.

The second stage runs the publication's measurement (`inv012/peak_trough.py`) on AndroidAPS and Trio records under the published event rules. An earlier draft had this stage decide the default level of the link to TDD. The simulation with a measured truth has since covered 100 to 250 mg/dL, and the measurement sat too far from the truth there to set one. What remains for the simulation is 300 mg/dL and a truth measured inside the free-living runs.

The third stage is simulation in TimSim, with runs paired by seed and an arm for unannounced meals. The candidate is also run with a mis-set profile ISF, and the TDD arm with a mis-set adjustment factor. The candidate is run with Trio's default limits and without them, and with the anchor at 140 and at 180 mg/dL. A harm rule is written before the runs and has two sides, time below 54 mg/dL and time above 250 mg/dL. Both are needed. For the worked person this model gives less insulin than AndroidAPS at 100% below 211 mg/dL and more above it, and at 380 mg/dL it asks for 3.33 times the profile's insulin.

The fourth stage is shadow running. Both values are calculated on the device each cycle and stored, and the existing one doses. That gives the distribution of the ratio between them for each person, and for the TDD arm a starting adjustment factor.

The fifth stage is use, on the developer's own device first and on released lines after that. It follows only when the third and fourth stages have passed, and it stops on the harm rule.

The first two stages are analysis and carry the analysis checklist. The last two change code that can change a dose. They carry unit and guard tests, the repository's review checklist, a replay against real history and a fresh-context review of the diff.

The way back at every stage is the mode selector. The existing modes stay in the code unchanged and remain the default until the fifth stage has run, and any build that includes the new mode names the build before it.

What is not settled

Some of what is missing can only come from the authors. How the curve is read and the level of the TDD arm depend on four things the publication does not give: which glucose an event is filed under, how the reference basal was defined, which insulin action model gave the insulin on board, and what daily doses the people measured were using. The authors have not been asked. They have also not been asked whether an equation derived by a manufacturer from its own users' records may be built into open-source code.

The link is settled only as far as INV-015 reaches, which is between adults who gave their own corrections with no algorithm dosing. Whether the profile ISF is as good a scale for a person whose loop does most of the correcting has not been tested. The same test on loop records would need the algorithm's insulin counted with the correction. Whether the candidate should also carry a multiplier for recent dose against the person's own average, which is what the link to TDD adds within a person, is open as well.

The anchor is open. The 140 mg/dL used here is an initial proposal, and the publication's reference is 180 mg/dL. The two differ by a factor of 1.44 in every dose, and the third stage of the evaluation runs both.

The level of the TDD arm is open, as described above. A level set from entered settings at 140 mg/dL may not apply to open-source systems, since the measurement on Loop, Trio and oref0 records read higher at 200 mg/dL.

It is an open question whether the steep part between 100 and 150 mg/dL should be used for dosing at all, given how many events there involved no net insulin. The question of whether to use the published quadratic below 100 mg/dL belongs with it.

The fall above 200 mg/dL is open in the same way. By the ratio of fall to insulin used, neither REPLACE-BG nor Loop reproduces it, whichever way that ratio is summarised. By the extra fall per extra unit, both do.

Parts of the existing systems were not read. The bolus calculator's sensitivity in dynamic mode is one, and the behaviour of autosens in that mode is another. TimSim's oref adapter has not been checked for whether it runs the logarithmic model closely enough to act as the comparator in the third stage.

The third stage has a further limit. In TimSim a unit of insulin does more at higher glucose, and the equation assumes it does less. A result from that stage describes the equation in a body built the other way, and whether people resemble the simulator in this respect is not known.

The last question is for the maintainers of AndroidAPS and Trio, and it is whether they want one shared definition. The two systems run the same equation and, at their defaults, return values 42% apart for the worked person at 200 mg/dL. A reference implementation with shared test vectors would be new to both.

Reproducing the numbers

`python3 -m inv013.candidate` writes `results/inv013_comparator.md`, which holds the numbers in this document's worked examples and in every table except the refit. `python3 -m inv013.refit` writes `results/inv013_refit.md`, with the refit tables and the refitted coefficients. It reads the per-person bins that `python3 -m inv012.peak_trough` writes, a file too large to keep in git, and checks them against the committed INV-012 table before fitting. `python -m inv014.known_truth --timsim <TimSim checkout>` writes `results/inv014_known_truth.md`, the simulation with a measured truth. It needs an interpreter with simglucose and took about three hours on seven workers. Its per-person tables stay out of git.
`python3 -m inv015.link` writes `results/inv015_link.md`, the comparison of the two links on REPLACE-BG corrections, and `inv015/tests/test_link.py` checks that the regression returns a link it was given.
`python3 -m inv016.replay --replay <INV-008 replay cache> --cohort <canonical_cohort.json>` writes `results/inv016_replay.md`, the replay on recorded histories. `inv013/tests/test_candidate.py` checks the logarithmic equation against the replay code of INV-008, Trio's form against AndroidAPS's, and the published fit against the values in the abstract and on the poster's figure. It also checks the ways of reading the curve, and that the refit returns a quartic it was given.

Figures from other analyses are quoted from where they were produced. The measurement on other cohorts is in `results/inv012_peak_trough.md` and section 7 of document 5. The adult figure of 1915 is from `docs/preprint-dosing-constants.md`, and the overnight test is in `oref-investigations/results/bridge_diabeloop.md`.

Earlier drafts of this document are in git. The first (commit fd3662e) proposed the profile-anchored, flat form now kept as an arm. The second (commit b324528) opened on the comparison with AndroidAPS and described both systems for a general reader. The third (commit 1667e8f) carried the link to TDD as the candidate and judged the equation against the simulations. The fourth (commit b19e3f1) anchored both links at 180 mg/dL.

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